

Robbie Hao, Master

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SUMMARY

M.S. Statistics candidate with applied machine learning experience building scalable predictive pipelines and evaluation workflows on large datasets. Developed Python factor research scripts, and analytical reports, evaluating 101 alpha signals across ~5,000 equities and 12 years of market data. Skilled in statistical modeling, time-series analysis, and factor research, with experience working with financial data to evaluate signals and portfolio performance.

EDUCATION

M.S. in Statistics—Duke University

Aug. 2025–Present

B.A. in Economics and Statistics—DUFE

Sept. 2021–Jul. 2025

WORK EXPERIENCE

Founder Fubon Fund

Jan.–Mar. 2024; Jul.–Sept. 2024

Quantitative Research Intern

- Built Python scripts to implement and test 101 Alpha-style factors across ~5,000 equities and 12 years of market data, supporting systematic signal evaluation and factor research.
- Screened and compared candidate factors using IC/ICIR analysis, rank-based portfolio tests, correlation checks, Lasso, and Random Forest selection to identify stronger and less redundant signals.
- Prepared research reports for senior analysts summarizing factor performance, signal stability, risk-return behavior, turnover, and out-of-sample results.
- Applied econometric frameworks (Fama-MacBeth, Fama-French) to evaluate statistical significance, exposure overlap, and robustness of factor construction.
- Built reusable Python scripts for factor orthogonalization, regression testing, and robustness checks, making repeated factor evaluation faster and more consistent.

RESEARCH

Graduate Researcher—DUKE

May 2024 – May 2025

- Developed a cost-conditioned reinforcement-learning agent for optimal trade execution in the ABIDES-Gym limit-order-book simulator, using observed spread, depth, volatility, and slippage to adapt trading speed.
- Built a controlled regime framework that varies liquidity-cost and timing-risk axes independently to test out-of-distribution robustness, benchmarking TWAP, Almgren-Chriss across baseline, thin, and unseen severe-thin markets.

Research Assistant—The Second Hospital of Dalian Medical University

May 2024–May 2025

- Co-developed a Python-based multi-model machine learning framework for automated feature selection and diagnostic prediction, integrating 9 algorithms (XGBoost, SVM, Random Forest, etc.).
- Applied the framework to high-dimensional lipidomics data with 498 features and 140 samples, identifying 10 statistically and clinically meaningful biomarkers for DIC risk diagnosis and achieving model AUC to ~0.95.

LEADERSHIP & SCHOOL ACTIVITIES

President, Photography Association—DUFE

2023–2024

- Partnered with Nikon to lead a photography competition with ~300 participants, highlighting hometown culture and raising funds for underserved communities.

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy, SciPy, Matplotlib), R, SQL.

ML/Stats: Random Forest, LightGBM, XGBoost, Lasso/Ridge Regression, Scikit-Learn, TensorFlow, Deep Learning methods, Reinforcement Learning, Bayesian methods

Finance: multi-factor equity models, backtesting, factor analyzing

Data & Tools: Git, Linux, Excel